

第二节

偏 导 数

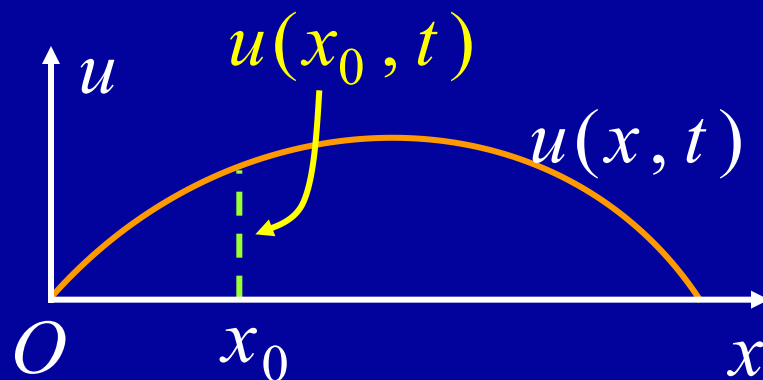
一、偏导数概念及其计算

二、高阶偏导数



一、偏导数定义及其算法

引例：研究弦在点 x_0 处的振动速度与加速度，就是将振幅 $u(x, t)$ 中的 x 固定于 x_0 处，求 $u(x_0, t)$ 关于 t 的一阶导数与二阶导数.



定义1. 设函数 $z = f(x, y)$ 在点 (x_0, y_0) 的某邻域内

极限
$$\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}$$

存在, 则称此极限为函数 $z = f(x, y)$ 在点 (x_0, y_0) 对 x

的偏导数, 记为 $\left. \frac{\partial z}{\partial x} \right|_{(x_0, y_0)}$; $\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)}$; $z_x|_{(x_0, y_0)}$;

$f_x(x_0, y_0)$; $f'_1(x_0, y_0)$.

注意:
$$f_x(x_0, y_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}$$

$$= \frac{d}{dx} f(x, y_0) \Big|_{x=x_0}$$



同样可定义对 y 的偏导数

$$\begin{aligned} f_y(x_0, y_0) &= \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y} \\ &= \frac{d}{dy} f(x_0, y) \Big|_{y=y_0} \end{aligned}$$

若函数 $z = f(x, y)$ 在域 D 内每一点 (x, y) 处对 x 或 y 偏导数存在, 则该偏导数称为偏导函数, 也简称为

偏导数, 记为 $\frac{\partial z}{\partial x}, \frac{\partial f}{\partial x}, z_x, f_x(x, y), f'_1(x, y)$

$\frac{\partial z}{\partial y}, \frac{\partial f}{\partial y}, z_y, f_y(x, y), f'_2(x, y)$



注意：偏导数的记号是一个整体记号，不能看作分子与分母之商。

偏导数的概念可以推广到二元以上的函数。

例如，三元函数 $u = f(x, y, z)$ 在点 (x, y, z) 处对 x 的偏导数定义为

$$f_x(x, y, z) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y, z) - f(x, y, z)}{\Delta x}$$

$$f_y(x, y, z) = ?$$

(请自己写出)

$$f_z(x, y, z) = ?$$



二元函数偏导数的几何意义:

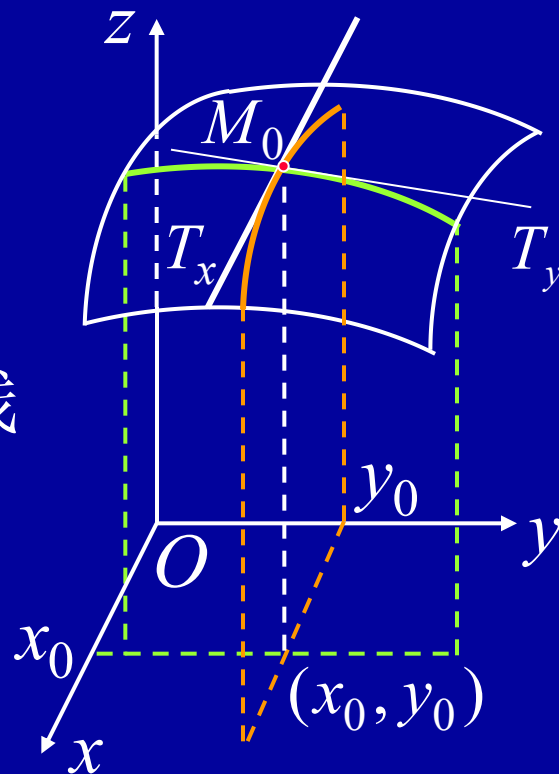
$$\left. \frac{\partial f}{\partial x} \right|_{\substack{x=x_0 \\ y=y_0}} = \left. \frac{d}{dx} f(x, y_0) \right|_{x=x_0}$$

是曲线 $\begin{cases} z = f(x, y) \\ y = y_0 \end{cases}$ 在点 M_0 处的切线

M_0T_x 对 x 轴的斜率.

$$\left. \frac{\partial f}{\partial y} \right|_{\substack{x=x_0 \\ y=y_0}} = \left. \frac{d}{dy} f(x_0, y) \right|_{y=y_0}$$

是曲线 $\begin{cases} z = f(x, y) \\ x = x_0 \end{cases}$ 在点 M_0 处的切线 M_0T_y 对 y 轴的斜率.



注意: 函数在某点各偏导数都存在,
但在该点**不一定连续**.

例如,
$$z = f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & x^2 + y^2 \neq 0 \\ 0 & , \quad x^2 + y^2 = 0 \end{cases}$$

显然
$$f_x(0, 0) = \frac{d}{dx} f(x, 0) \Big|_{x=0} = 0$$

$$f_y(0, 0) = \frac{d}{dy} f(0, y) \Big|_{y=0} = 0$$

在上节已证 $f(x, y)$ 在点 $(0, 0)$ 并不连续!



例1. 求 $z = x^2 + 3xy + y^2$ 在点 $(1, 2)$ 处的偏导数.

解法1 $\frac{\partial z}{\partial x} = 2x + 3y, \quad \frac{\partial z}{\partial y} = 3x + 2y$

先求后代

$$\therefore \left. \frac{\partial z}{\partial x} \right|_{(1,2)} = 2 \cdot 1 + 3 \cdot 2 = 8, \quad \left. \frac{\partial z}{\partial y} \right|_{(1,2)} = 3 \cdot 1 + 2 \cdot 2 = 7$$

解法2 $z|_{y=2} = x^2 + 6x + 4$

先代后求

$$\left. \frac{\partial z}{\partial x} \right|_{(1,2)} = (2x + 6) \Big|_{x=1} = 8$$

$$z|_{x=1} = 1 + 3y + y^2$$

$$\left. \frac{\partial z}{\partial y} \right|_{(1,2)} = (3 + 2y) \Big|_{y=2} = 7$$



例2. 设 $z = x^y$ ($x > 0$, 且 $x \neq 1$), 求证

$$\frac{x}{y} \frac{\partial z}{\partial x} + \frac{1}{\ln x} \frac{\partial z}{\partial y} = 2z$$

证: $\frac{\partial z}{\partial x} = yx^{y-1}, \quad \frac{\partial z}{\partial y} = x^y \ln x$

$$\therefore \frac{x}{y} \frac{\partial z}{\partial x} + \frac{1}{\ln x} \frac{\partial z}{\partial y} = x^y + x^y = 2z$$

例3. 求 $r = \sqrt{x^2 + y^2 + z^2}$ 的偏导数.

解: $\frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2 + y^2 + z^2}} = \frac{x}{r}$

$$\frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$



例4. 已知理想气体的状态方程 $pV = RT$ (R 为常数),

求证: $\frac{\partial p}{\partial V} \cdot \frac{\partial V}{\partial T} \cdot \frac{\partial T}{\partial p} = -1$

证: $p = \frac{RT}{V}, \quad \frac{\partial p}{\partial V} = -\frac{RT}{V^2}$

$$V = \frac{RT}{p}, \quad \frac{\partial V}{\partial T} = \frac{R}{p}$$

$$T = \frac{pV}{R}, \quad \frac{\partial T}{\partial p} = \frac{V}{R}$$

$$\therefore \frac{\partial p}{\partial V} \cdot \frac{\partial V}{\partial T} \cdot \frac{\partial T}{\partial p} = -\frac{RT}{pV} = -1$$

说明: 此例表明,
偏导数记号是一个
整体记号, 不能看作
分子与分母的商!



二、高阶偏导数

设 $z = f(x, y)$ 在域 D 内存在连续的偏导数

$$\frac{\partial z}{\partial x} = f_x(x, y), \quad \frac{\partial z}{\partial y} = f_y(x, y)$$

若这两个偏导数仍存在偏导数, 则称它们是 $z = f(x, y)$ 的二阶偏导数. 按求导顺序不同, 有下列四个二阶偏导数:

$$\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial^2 z}{\partial x^2} = f_{xx}(x, y); \quad \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial^2 z}{\partial x \partial y} = f_{xy}(x, y)$$

$$\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial^2 z}{\partial y \partial x} = f_{yx}(x, y); \quad \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial^2 z}{\partial y^2} = f_{yy}(x, y)$$



类似可以定义更高阶的偏导数.

例如, $z = f(x, y)$ 关于 x 的三阶偏导数为

$$\frac{\partial}{\partial x} \left(\frac{\partial^2 z}{\partial x^2} \right) = \frac{\partial^3 z}{\partial x^3}$$

$z = f(x, y)$ 关于 x 的 $n-1$ 阶偏导数, 再关于 y 的一阶偏导数为

$$\frac{\partial}{\partial y} \left(\frac{\partial^{n-1} z}{\partial x^{n-1}} \right) = \frac{\partial^n z}{\partial x^{n-1} \partial y}$$



例5. 求函数 $z = e^{x+2y}$ 的二阶偏导数及 $\frac{\partial^3 z}{\partial y \partial x^2}$.

解： $\frac{\partial z}{\partial x} = e^{x+2y}$

$$\frac{\partial z}{\partial y} = 2e^{x+2y}$$

$$\frac{\partial^2 z}{\partial x^2} = e^{x+2y}$$

$$\frac{\partial^2 z}{\partial x \partial y} = 2e^{x+2y}$$

$$\frac{\partial^2 z}{\partial y \partial x} = 2e^{x+2y}$$

$$\frac{\partial^2 z}{\partial y^2} = 4e^{x+2y}$$

$$\frac{\partial^3 z}{\partial y \partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial^2 z}{\partial y \partial x} \right) = 2e^{x+2y}$$

注意：此处 $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$, 但这一结论并不总成立.



例如, $f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2}, & x^2 + y^2 \neq 0 \\ 0, & x^2 + y^2 = 0 \end{cases}$

$$f_x(x, y) = \begin{cases} y \frac{x^4 + 4x^2y^2 - y^4}{(x^2 + y^2)^2}, & x^2 + y^2 \neq 0 \\ 0, & x^2 + y^2 = 0 \end{cases}$$

$$f_y(x, y) = \begin{cases} x \frac{x^4 - 4x^2y^2 - y^4}{(x^2 + y^2)^2}, & x^2 + y^2 \neq 0 \\ 0, & x^2 + y^2 = 0 \end{cases}$$

$$\left. \begin{aligned} f_{xy}(0,0) &= \lim_{\Delta y \rightarrow 0} \frac{f_x(0, \Delta y) - f_x(0, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{-\Delta y}{\Delta y} = -1 \\ f_{yx}(0,0) &= \lim_{\Delta x \rightarrow 0} \frac{f_y(\Delta x, 0) - f_y(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x} = 1 \end{aligned} \right\} \text{二者不等}$$



例6. 证明函数 $u = \frac{1}{r}$, $r = \sqrt{x^2 + y^2 + z^2}$ 满足拉普拉斯

方程 $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

证: $\frac{\partial u}{\partial x} = -\frac{1}{r^2} \frac{\partial r}{\partial x} = -\frac{1}{r^2} \cdot \frac{x}{r}$

$$\frac{\partial^2 u}{\partial x^2} = -\frac{1}{r^3} + \frac{3x}{r^4} \cdot \frac{\partial r}{\partial x} = -\frac{1}{r^3} + \frac{3x^2}{r^5}$$

利用对称性, 有 $\frac{\partial^2 u}{\partial y^2} = -\frac{1}{r^3} + \frac{3y^2}{r^5}$, $\frac{\partial^2 u}{\partial z^2} = -\frac{1}{r^3} + \frac{3z^2}{r^5}$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = -\frac{3}{r^3} + \frac{3(x^2 + y^2 + z^2)}{r^5} = 0$$



定理. 若 $f_{xy}(x,y)$ 和 $f_{yx}(x,y)$ 都在点 (x_0, y_0) 连续, 则

$$f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0) \quad (\text{证明略})$$

本定理对 n 元函数的高阶混合导数也成立.

例如, 对三元函数 $u = f(x, y, z)$, 当三阶混合偏导数在点 (x, y, z) 连续时, 有

$$\begin{aligned} f_{xyz}(x, y, z) &= f_{yzx}(x, y, z) = f_{zxy}(x, y, z) \\ &= f_{xzy}(x, y, z) = f_{yxz}(x, y, z) = f_{zyx}(x, y, z) \end{aligned}$$

说明: 因为初等函数的偏导数仍为初等函数, 而初等函数在其定义区域内是连续的, 故求初等函数的高阶导数可以选择方便的求导顺序.



定理. 若 $f_{xy}(x,y)$ 和 $f_{yx}(x,y)$ 都在点 (x_0, y_0) 连续, 则

$$f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$$

证: 令 $F(\Delta x, \Delta y) = f(\underline{x_0 + \Delta x}, y_0 + \Delta y) - f(\underline{x_0 + \Delta x}, y_0)$
 $\quad \quad \quad - f(\underline{x_0}, y_0 + \Delta y) + f(\underline{x_0}, y_0)$

又令 $\varphi(x) = f(x, y_0 + \Delta y) - f(x, y_0)$

$$\psi(y) = f(x_0 + \Delta x, y) - f(x_0, y)$$

则 $F(\Delta x, \Delta y) = \varphi(x_0 + \Delta x) - \varphi(x_0)$

$$= \varphi'(x_0 + \theta_1 \Delta x) \Delta x \quad (0 < \theta_1 < 1)$$

$$= [f_x(x_0 + \theta_1 \Delta x, \underline{y_0 + \Delta y}) - f_x(x_0 + \theta_1 \Delta x, \underline{y_0})] \Delta x$$

$$= f_{xy}(x_0 + \theta_1 \Delta x, y_0 + \theta_2 \Delta y) \Delta x \Delta y \quad (0 < \theta_1, \theta_2 < 1)$$



同样

$$\begin{aligned} F(\Delta x, \Delta y) &= f(x_0 + \Delta x, \underline{y_0 + \Delta y}) - f(x_0 + \Delta x, \underline{y_0}) \\ &\quad - f(x_0, \underline{y_0 + \Delta y}) + f(x_0, \underline{y_0}) \\ &= \psi(y_0 + \Delta y) - \psi(y_0) \\ &= f_{yx}(x_0 + \theta_3 \Delta x, y_0 + \theta_4 \Delta y) \Delta x \Delta y \\ &\qquad\qquad\qquad (0 < \theta_3, \theta_4 < 1) \end{aligned}$$

$$\begin{aligned} \therefore f_{xy}(x_0 + \theta_1 \Delta x, y_0 + \theta_2 \Delta y) \\ = f_{yx}(x_0 + \theta_3 \Delta x, y_0 + \theta_4 \Delta y) \end{aligned}$$

因 $f_{xy}(x, y), f_{yx}(x, y)$ 在点 (x_0, y_0) 连续, 故令 $\Delta x \rightarrow 0$, $\Delta y \rightarrow 0$ 得 $f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$



内容小结

1. 偏导数的概念及有关结论

- 定义; 记号; 几何意义
- 函数在一点偏导数存在 $\Rightarrow$ 函数在此点连续
- 混合偏导数连续 $\Rightarrow$ 与求导顺序无关

2. 偏导数的计算方法

- 求一点处偏导数的方法 $\left\{ \begin{array}{l} \text{先代后求} \\ \text{先求后代} \\ \text{利用定义} \end{array} \right.$
- 求高阶偏导数的方法 —— 逐次求导法

(与求导顺序无关时, 应选择方便的求导顺序)



思考与练习

P129 题 5, 6

解答提示: P129 题 5

当 $x^2 + y^2 \neq 0$ 时,

$$f_x(x, y) = \frac{\partial}{\partial x} \left(\frac{x^2 y}{x^2 + y^2} \right) = \frac{2xy^3}{(x^2 + y^2)^2}$$

$$f_y(x, y) = \frac{\partial}{\partial y} \left(\frac{x^2 y}{x^2 + y^2} \right) = \frac{x^2(x^2 - y^2)}{(x^2 + y^2)^2}$$

当 $x^2 + y^2 = 0$, 即 $x=y=0$ 时,

$$f_x(0,0) = \left. \frac{d}{dx} f(x,0) \right|_{x=0} = 0$$

$$f_y(0,0) = \left. \frac{d}{dy} f(0,y) \right|_{y=0} = 0$$



P129 题6

$$(1) \quad \frac{\partial z}{\partial x} = \frac{1}{x+y^2}, \quad \frac{\partial z}{\partial y} = \frac{2y}{x+y^2}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{-1}{(x+y^2)^2}, \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{-2y}{(x+y^2)^2}, \quad \frac{\partial^2 z}{\partial y^2} = \frac{2(x-y^2)}{(x+y^2)^2}$$

$$(2) \quad \frac{\partial z}{\partial x} = yx^{y-1}, \quad \frac{\partial z}{\partial y} = x^y \ln x$$

$$\frac{\partial^2 z}{\partial x^2} = y(y-1)x^{y-2}, \quad \frac{\partial^2 z}{\partial x \partial y} = x^{y-1} + yx^{y-1} \ln x$$

$$\frac{\partial^2 z}{\partial y^2} = x^y \ln^2 x$$



作业

P68 1 (4) , (6) , (8) ; 3; 5;

6 (3) ; 7; 8; 9 (2)



备用题 设 $z = f(u)$, 方程 $u = \varphi(u) + \int_y^x p(t) dt$

确定 u 是 x, y 的函数, 其中 $f(u), \varphi(u)$ 可微, $p(t), \varphi'(u)$ 连续, 且 $\varphi'(u) \neq 1$, 求 $p(y) \frac{\partial z}{\partial x} + p(x) \frac{\partial z}{\partial y}$.

解: $\frac{\partial z}{\partial x} = f'(u) \frac{\partial u}{\partial x}, \quad \frac{\partial z}{\partial y} = f'(u) \frac{\partial u}{\partial y}$

$$\left. \begin{aligned} \frac{\partial u}{\partial x} &= \varphi'(u) \frac{\partial u}{\partial x} + p(x) \\ \frac{\partial u}{\partial y} &= \varphi'(u) \frac{\partial u}{\partial y} - p(y) \end{aligned} \right\} \Rightarrow \begin{cases} \frac{\partial u}{\partial x} = \frac{p(x)}{1 - \varphi'(u)} \\ \frac{\partial u}{\partial y} = \frac{-p(y)}{1 - \varphi'(u)} \end{cases}$$

$$\therefore p(y) \frac{\partial z}{\partial x} + p(x) \frac{\partial z}{\partial y} = f'(u) \left[p(y) \frac{\partial u}{\partial x} + p(x) \frac{\partial u}{\partial y} \right] = 0$$

